

HACKATHON SUBMISSION DECK

SmartCharge AI

Intelligent EV Load Balancing System

Built for **Lablab.ai IBM Bob Hackathon 2026** • Powered by **watsonx.ai**

The Crisis of "Dumb" EV Charging

Residential grids are under unprecedented strain due to uncoordinated high-capacity electrical charging inputs.

- ⚡ **Peak Demand Congestion:** Millions of EV owners plug in simultaneously at 6:00 PM upon returning from work, threatening localized grid brownouts.
- ⚡ **Time-of-Use Financial Tariffs:** Blindly drawing power during evening demand peaks exposes consumers to pricing spikes (\$0.35+/kWh).
- ⚡ **Renewable Disconnect:** Local home-solar arrays often feed excess green energy back into the grid for minimal return during midday hours instead of capturing it.

\$500 Annual Loss

The average electric vehicle owner completely wastes up to \$500 a year by charging during peak-hour utility premiums rather than shifting to off-peak pricing windows.

Grid Inefficiency

Surge peaks force utilities to rely heavily on fossil-fuel burning peaker plants, raising the carbon footprint of otherwise clean transportation systems.

The Solution: Autonomous Energy Brokerage

SmartCharge AI acts as an autonomous home broker, evaluating power constraints every 5 seconds to dynamically choose optimal states.

Explainable Intelligence Loop

By parsing environment vectors against cost and capacity rules, the logic yields contextual optimization with clear natural-language justifications for the owner.

 **FAST_CHARGE Mode:** Engaged during premium solar generation or ultra-low off-peak grid pricing to capture massive energy volume.

 **ECO_MODE Mode:** Executes low-impact balancing during normal utility windows, trickle-charging the vehicle to protect battery chemistry.

 **PAUSED Mode:** Entirely halts the drawing loop when grid prices reach extreme peak tariffs, shielding users from financial spikes.

40%

COST REDUCTION

100%

GREEN CAPTURE

A Modern, Decoupled Architecture

Our production architecture uses five distinct micro-layers communicating smoothly over secure isolated networks.

1. Ingestion Layer

Telemetry Simulator: Independent Python service generating pricing maps, solar generation metrics, and vehicle SoC metrics every 5 seconds.

Python

HTTP POST

2. Processing & Storage

FastAPI & Persistence: High-performance async router that ingests telemetry payloads, orchestrates cache entries via Redis, and stores tracking records in PostgreSQL / SQLite.

FastAPI

SQLAlchemy

3. Frontend Analytics

React UI Dashboard: Dynamic real-time user interface utilizing Recharts time-series arrays, responsive SVG gauges, and manual command injection overrides.

React 18

Tailwind CSS

Accelerating Production with IBM Bob

IBM Bob acted as our core pair-programming engineer, enabling us to design, validate, and launch the multi-service pipeline in record time.

- ⚡ **Data Optimization:** Constructed clean data schemas mapping unflattened JSON payloads safely into backend models.
- ⚡ **Time-Series Visualization:** Structured responsive array handlers for Recharts rendering loops, resolving timestamp duplication errors.
- ⚡ **Infrastructure Routing:** Diagnosed and resolved complex Docker deployment networking, and Vite proxy cross-origin restrictions.

Exported Bob IDE Session Verification

Full session summaries containing our collaborative task paths are compiled directly into the repository:

```
smartcharge-ai/ └─ ibm-bob-reports/ └─  
  bob_task_may-15-2026_9-13-20-pm.md └─  
  bob_task_may-15-2026_10-05-53-pm.md └─  
  bob_task_may-15-2026_10-32-30-pm.md
```

Live Deployment Infrastructure Architecture

The full architecture is deployed bare-metal across cloud containers on Railway, using strict secure internal communications.

Secure Cloud Networking Matrix

Services interact smoothly using optimized environment vectors:

Service	Endpoint Path Variable	Protocol Type
Frontend UI	smartcharge-ai-demo.up.railway.app	Public Web
FastAPI Backend	smartcharge-ai-ibm-bob-production.up.railway.app	API Docs Live
Simulator Worker	http://backend.railway.internal	Secure Private

✓ Passed

VITE SECURITY HOST MAPPING

0.0.0.0

CONTAINER ACCESS GATEWAY

Active

REDIS CACHE CONNECTION

Human-in-the-Loop: Instant Manual Override

Energy automation demands human priority. SmartCharge AI guarantees immediate user agency over optimization loops.

- ⚡ **Optimistic State Snapping:** UI adapts instantly to manual control clicks, bypassing typical backend network polling lag.
- ⚡ **15-Second AI Safety Lockout:** Clicking manual overrides suspends the autonomous AI load balancing temporarily, keeping the system locked to the user's specific choice.
- ⚡ **Database Injection Tracking:** Injects custom programmatic override data vectors into SQL history with 100% confidence markers.

Live Override Ingestion Signature

When an override button is pressed, a priority transaction bypasses the autonomous model:

```
{ "request_id": "manual-4b7e2a8c", "charging_mode":  
  "FAST_CHARGE", "reasoning": "👤 USER OVERRIDE: AI  
  suspended. Forcing fast charge.", "confidence":  
  1.0, "overridden": true }
```

Market Commercialization & Monetization Blueprint

| B2C Subscription Software model

Direct-to-consumer software-as-a-service application matching EV residential homes.

- ⚡ **Premium Tier:** \$9.99/month subscription offering continuous optimization metrics.
- ⚡ **Battery Preservation:** Proprietary trickle algorithms extending expensive car battery health by up to 25%.

| B2B Utility Grid Partnerships

Aggregated regional charging infrastructure acting as virtual power plants (VPPs).

- 🌱 **Demand Response:** Power utility providers incentivize mass users to automatically pause loops during grid emergencies.
- 🌱 **Carbon Balancing:** Aggregated carbon certificate token generation for industrial ESG balancing offsets.